

Bonus - Grand Challenge of Safety Critical Human Motion Generation

Team Members:

- **Prajwal Kumar**
- **Jeet Vijaywargi**
- **Hannah Manheimer**

Team Contributions

In accordance with the honor code, we acknowledge the following individual contributions to this project:

- **Jeet Vijaywargi:** Responsible for Part 1.1 (Run the SMPLH data rendering). Jeet managed the initial AWS environment setup, installed the necessary visualization dependencies (Open3D, Pyrender), and successfully executed the baseline rendering script to validate the SMPL-H model visualization pipeline.
- **Hannah Manheimer:** Responsible for Part 2.1 (Run the skeleton controller). Hannah set up the CARLA simulation environment for the skeleton controller, managed the connection between the Python client and the CARLA server, and generated the baseline motion video to establish the initial performance of the walker blueprint.
- **Prajwal Kumar:** Responsible for Part 1.2, Part 1.3, and Part 2.2. I executed the WHAM model pipeline on Google Colab, resolving critical library version mismatches (PyTorch/MKL) to generate the SMPL data. I then modified the `render_smplh.py` script to parse this custom data on AWS. Finally, I derived and implemented the coordinate transformation matrices in `joint_mapping.py` to correct the skeletal misalignment, ensuring smooth human motion in the CARLA simulation.

Introduction

Simulating realistic human motion is critical for evaluating the safety of autonomous systems. This project focuses on the end-to-end pipeline of estimating human motion from monocular video using State-of-the-Art (SOTA) computer vision techniques and integrating that motion into the CARLA simulator for safety-critical testing. The project is divided into two phases: motion extraction using the WHAM (World-grounded Humans with Accurate Motion) model and skeleton control within the simulation environment.

1. From Video to SMPLH

In this section, we utilized the WHAM framework to recover global human motion from a video sequence and visualized the resulting SMPL-H mesh.

1.1 Run the SMPLH Data Rendering

We first established a baseline rendering pipeline on the AWS instance. Using the provided `render_smplh.py` script and the HumanEva dataset, we verified the environment's capability to render 3D human meshes in a headless configuration (using `xvfb`).

- **Deliverable:** <https://drive.google.com/file/d/1-cecgFyhP9h7dpbUrbu6584tFxdAQNi/view?usp=sharing>

1.2 Run the Model to Collect SMPLH Data

We utilized Google Colab to run the WHAM model on the demo video (`IMG_9732.mov`).

- **Technical Challenge:** The default Colab environment introduced a library mismatch between PyTorch 1.11 and the latest Intel MKL (2025.x), causing an undefined symbol: `iJIT_NotifyEvent` error.
- **Solution:** We resolved this by creating a fresh Conda environment and explicitly pinning `mkl=2021.4.0` and `intel-openmp=2021.4.0` before installing PyTorch.
- **Execution:** The model was executed with the `--save_pkl` flag to extract the pose parameters and `--visualize` to generate the mesh overlay.
- **Deliverable (Data):**
https://drive.google.com/file/d/18H4vBxEk5On_yUsgrwbo1aEb_EA1XCMP/view?usp=sharing
- **Deliverable (Mesh Video):**
https://drive.google.com/file/d/1kf1hWi8ESjzu9B5_wk_qfHMWgrWf2638/view?usp=sharing

1.3 Render the Saved SMPLH Data

We transferred the generated `wham_output.pkl` to the AWS instance and modified the rendering script to visualize our specific motion data.

- **Code Modifications:** The `render_smplh.py` script was updated to:
 1. Parse the nested dictionary structure of the WHAM output `.pkl` file.

2. Correctly extract the pose (72 parameters) and betas (shape parameters).
 3. Fix a dimension mismatch error where the shape parameters were being loaded as (N, 10) instead of (10,).
- **Deliverable:** https://drive.google.com/file/d/1x14muTCg-gPITzb_yqygSc1KrrMmj5H/view?usp=sharing

2. From SMPLH to CARLA

This section focuses on retargeting the extracted SMPL motion onto a CARLA walker skeleton to enable realistic pedestrian simulation.

2.1 Run the Skeleton Controller

We ran the provided starter code (test_controller.py) to map the motion onto a CARLA pedestrian. The initial result showed significant skeletal distortion due to coordinate system mismatches between the SMPL model (Z-up) and the CARLA/Unreal Engine walker skeleton (Y-up/different joint axes).

- **Deliverable:** <https://drive.google.com/file/d/1tKwZ3lpAA7ryxjepWNJv6-jC6mL28G6s/view?usp=sharing>

2.2 Modify the Skeleton Controller to Render Motion Correctly

To correct the "broken" motion seen in Part 2.1, we analyzed the coordinate systems and implemented a custom mapping_matrix in joint_mapping.py.

Transformation Logic:

1. Global Root Adjustment: SMPL uses a Z-up coordinate system, while the target walker in CARLA required a -90° rotation around the X-axis to orient the root upright.
2. Arm Alignment: SMPL defines zero-rotation as a T-Pose. The CARLA skeleton's local bone axes differed, requiring:
 - Left Arm: +90° rotation around the Z-axis for the Shoulder and Elbow.
 - Right Arm: -90° rotation around the Z-axis for the Shoulder and Elbow.
3. Implementation: We replaced the complex manual calculations in joint_mapping.py with a clean dictionary of rotation matrices using scipy.spatial.transform.

Code Snippet (Mapping Function):

```

def get_mapping_matrix():
    mapping = {}
    # Root Adjustment: Rotate -90 degrees X (Fix Z-up to Y-up)
    r_root = R.from_euler('x', -90, degrees=True).as_matrix()
    mapping[0] = r_root
    # Arm Adjustments: Align T-pose axes
    r_l_arm = R.from_euler('z', 90, degrees=True).as_matrix() # Left Side
    r_r_arm = R.from_euler('z', -90, degrees=True).as_matrix() # Right Side
    # Apply to Shoulders (16/17), Elbows (18/19), and Wrists (20/21)
    for joint in [16, 18, 20]: mapping[joint] = r_l_arm
    for joint in [17, 19, 21]: mapping[joint] = r_r_arm

    return mapping

```

This mapping successfully rectified the skeletal pose, resulting in smooth, natural human motion within the simulator.

- **Deliverable (Mapping Code):**
https://drive.google.com/file/d/1TyDh5myJhbf_oKDnyvo8U8yEjzcCiQR_/view?usp=sharing
- **Deliverable (Corrected Video):**
<https://drive.google.com/file/d/1uA5MdQztnnbWm17BncwQaJhK-g7TzFjA/view?usp=sharing>